

Datasets, Metrics, Benchmarks and Future Research in Autonomous Driving: A Review

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Abstract—Data-driven autonomous driving is a hot topic in academic and industry research due to its impressive performance, flexible mobility, and reduced human intervention. However, the development of this technology relies heavily on large datasets that contain accurately annotated data, obtained through artificial or semi-automated strategies. Consequently, datasets play a crucial role in autonomous driving, and their characteristics significantly impact the effectiveness of algorithms. Currently, there are several diverse datasets available, such as KITTI and CityScape, that cover various tasks. However, researchers often overlook the unique features, similarities, and specificities of these datasets. Furthermore, to the best of our knowledge, there is a lack of survey articles focusing on special metrics and benchmark performance on different datasets in autonomous driving. Therefore, the purpose of this article is to analyze autonomous driving datasets, guide researchers on collecting and utilizing relevant datasets, summarize evaluation strategies, analyze benchmark performance, and provide future research points to enrich the autonomous driving community. We believe that this work

will assist researchers in evaluating their data using suitable metrics and offer a fresh perspective on autonomous driving.

Index Terms—Autonomous driving, benchmarks, dataset, metrics.

“It’s time for smart-sized, “data-centric” solutions to big issues” — Andrew Ng

I. INTRODUCTION

AUTONOMOUS driving has emerged as an effective technological solution to enhance transportation efficiency and reduce traffic risks in our daily lives. Data-driven approaches in autonomous driving have demonstrated impressive performance due to their availability, comprehensive explanation, and flexibility [1]–[3]. However, these methods heavily rely on large amounts of annotated data for training and testing. As a result, various professional datasets specific to autonomous driving have been collected and released by academic institutions and manufacturing companies, catering to different tasks. For instance, the Caltech Pedestrians dataset [4] offers benchmarks and evaluation tools for pedestrian detection tasks. Similarly, datasets like KITTI [5], Waymo [6], and nuScenes [7] have played significant roles in advancing scene understanding and object detection in autonomous driving. Moreover, the Oxford dataset [8] is commonly utilized for localization and mapping methods, allowing researchers to compare translation errors with ground truth obtained from the global navigation satellite system (GNSS). However, it can be challenging for researchers, especially beginners, to collect and navigate through the vast array of available datasets. This can lead to confusion and consume significant time. In this article, we aim to address this issue by providing a detailed introduction to these datasets. The structure of the article will encompass sections on datasets, metrics, benchmarks, and future research.

In this article, we summarize over 100 datasets related to autonomous driving and categorize them into three groups based on data type: real-world data, synthetic data, and simulation platforms. Real-world data is captured from vehicles or traffic sensors in real environments and is primarily used for tasks such as localization, perception, and tracking. Synthetic data, generated in virtual environments like 3D games or augmented reality, is typically employed for domain adaptation and data augmentation methods. Simulation platforms allow researchers to design novel environments and driving scenarios to safely validate end-to-end methods. Autonomous driving datasets may be suited to one or multiple tasks, and data collection may involve a single vehicle, multiple vehicles, or non-vehicle platforms. Sensors used include, but are not limited to, cameras, LiDAR, millimeter-wave radar, infrared

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cameras, inertial navigation systems, and CAN-BUS data. One of the classification configurations for the collection platform and the supported tasks is shown in Fig. 1.

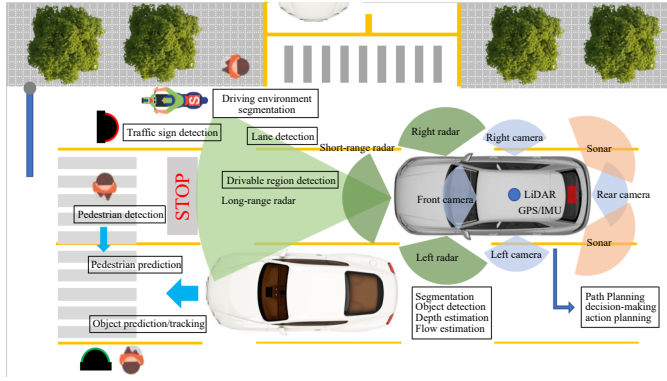


Fig. 1. One of the classification configurations for the collection platform with full field of view (FoV) cameras, one top LiDAR, multiple radars, sonars, and one GNSS (global positioning system) (GPS), inertial measurement unit (IMU) and the supported tasks such as various detection, segmentation tasks, planning and decision-making).

The definition and design of metrics are crucial aspects of dataset creation, yet they are often overlooked by readers and researchers. Metrics should align with the data distribution and hold mathematical significance for the collected data. The dataset creators establish baseline results, enabling external users to evaluate their models against standardized benchmarks. Metrics also reflect the characteristics and features of the data, thereby contributing to the dataset itself. This article introduces unique metrics developed specifically for autonomous driving scenarios or tasks and analyzes their intended purposes. In addition, we provide data to support benchmark evaluations and analyze the performance variations of state-of-the-art (SOTA) algorithms across various datasets, attempting to explain the underlying reasons for these differences.

Furthermore, we analyze existing challenges from the perspective of autonomous driving data and seek to identify key areas for future research, including the advancement of end-to-end autonomous driving, the standardization of autonomous driving data, and the impact of large language models and corresponding datasets on the development of autonomous driving technologies. We believe that fully autonomous driving will eventually become a reality; however, the journey toward this goal is bound to be complex and challenging. Along the way, new disciplines, innovative ideas, and novel algorithms may emerge and be integrated, driven by the collective efforts of many researchers. We hope that our work provides valuable insights from a data perspective, contributing to this ongoing development. We summarize the contributions of this article as:

- 1) We compile and analyze over 100 datasets, including real and virtual environments, as well as simulation platforms, providing detailed feature analysis, comparison, and special metrics in autonomous driving.
- 2) We provide benchmark data and analyze performance variations of state-of-the-art algorithms across datasets, exploring underlying reasons for these differences.

- 3) We highlight the current limitations of existing datasets and explore the potential impacts on the future development of autonomous driving influenced by emerging technologies such as end-to-end systems and large language models.

II. AUTONOMOUS DRIVING TASKS

Before we organize the autonomous driving datasets, here is a brief introduction to the tasks involved in autonomous driving as shown in Fig. 2. Typically, autonomous driving datasets are collected using sensors mounted on modified vehicles, autonomous robots, handheld platforms, drones, roadside infrastructure, and other platforms. These sensors include, but are not limited to, radar, cameras, GNSS, IMU, and HD maps. Depending on the collected data and subsequent manual processing, these datasets can support a variety of tasks, including perception, decision-making, and planning. Perception tasks—which constitute the majority of data coverage in autonomous driving—encompass segmentation (semantic segmentation, instance segmentation, panoptic segmentation, point segmentation), understanding (depth estimation, optical estimation, stereo matching, and scene understanding), detection (drivable region detection, sign detection, lane detection, tracking), prediction, and localization. Once an autonomous system completes these tasks, it can utilize destination information to perform path planning, decision-making, and control tasks. Additionally, datasets are available to support multi-vehicle cooperative tasks and analyses of driver behavior.

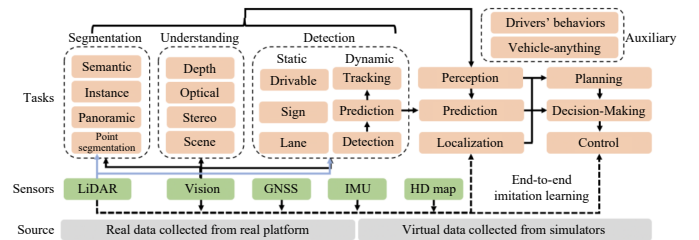


Fig. 2. The relationship of data source, sensors, and following tasks.

Most released autonomous dataset collection workflows follow a structured process as illustrated in Fig. 3. This process typically begins with the selection and definition of relevant scenes, where more challenging and complex scenes are prioritized for their greater contribution to the autonomous driving community. Data is generally collected either on-site or through virtual simulations, with the choice of platform and sensor models determined by the specific requirements of the dataset. The workflow then includes equipment installation, operating system and driver setup, sensor timestamp synchronization, parameter calibration, and data transfer validation, all according to a standardized blueprint. Once these initial

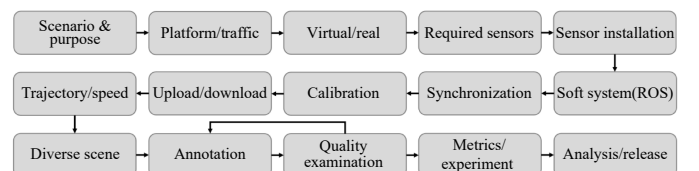


Fig. 3. A typical workflow for data collection in autonomous driving.

TABLE I
THE PLATFORM, TYPES OF SENSORS, SYNCHRONIZATION METHOD, AND COVERING TASKS FOR CRUCIAL DATASETS

Dataset	Platform	Sensor	Sync	Task
KITTI [5]	VW	1xHDL-64E LiDAR; 2xPointGrey Flea2 camera (color) 2xPointGrey Flea2 camera (grayscale); GPS/IMU	L2C	Lo 2-3DOD SS IS PP Dp Fl Sc St
A*3D [9]	—	1xHDL-64ES3; 2xPointGrey CM3-U3-31S4C-CS camera	ROS (s)	2D-3DOD
A2D2 [10]	Audi-Q7	5xVLP-16; 1xSekonix SF3325-100; 5xSekonix SF3324-100	L2C	Lo 2D-3DOD SS IS E2E
Argoverse [11]	Ford	2xroof-mounted LiDAR; 7xcameras (30 Hz); 2xstereo cameras (5 Hz); GPS; HD map	—	2-3DTK PD
BDD00K [12]	Multi	720 P images (30 Hz); GPS/IMU	—	SC 2-3DOD LD DA SS IS 2DTK STK E2E
D ² -City [13]	Multi	720/1080 P images (25 Hz)	—	2DOD 2DTK
H3D [14]	Honda	3xPointGrey camera (30 Hz); 1xHDL-64E (10 Hz); GNSS (100 Hz)	G2S	2-3DOD 2-3DTK
nuScenes [7]	2 Zoe	6xcolor camera (12 Hz); 1x32LiDAR (20 Hz); 5xradar (13 Hz); GPS/IMU (1K Hz); CAN; HD map	L2C	Lo 2-3DOD 2D-3DTK
ONCE [15]	—	1x40LiDAR (10 Hz); 7xcolor camera (12 Hz)	L2C	2-3DOD
PandaSet [16]	min VAN	1xHesai-Pandar64 (10 Hz); 1xHesai PandarGT (10 Hz); 5xLeopard LiColor camera (10 Hz); 1xLeopard LI long focus camera; GNSS/IMU	L2C	2-3DOD
Apollo [17]	mid SUV	2xVUX-1HA laser; 6xVMX-CS6 cameras; GNSS/IMU	—	SS IS PP
Waymo [6]	mid MPV	1xmid-range LiDAR (10 Hz); 4xshort-range LiDARs (10 Hz); 5xcolor cameras	—	2-3DOD SS IS PP
Oxford [8]	Nissan	1xBBX3-13S2C-38 (16 Hz); 3xGS2-FW-14S5C-C (11 Hz); 2xSICK LMS-151 2D LiDAR (50 Hz); 1xSICK LD-MRS 4-LiDAR; GNSS (50 Hz)	C2C	Lo
CityScape [18]	Car	2xOnSemi AR0331 (17 Hz)	C2C	SS IS
Lyft-Pre [19]	20 Cars	1x64-LiDAR (10 Hz); 2xbumper 40-LiDAR (10 Hz); 4xraders (10 Hz); 7xcolor cameras (10 Hz)	—	PD PL
Lyft-L5 [19]	Multi	3x40-LiDAR (10 Hz); 6xWFOV cameras (10 Hz); 1xLF camera (10 Hz); HD map	L2C	2D-3DOD
BLVD [20]	Car	1xVelodyne HDL-64LiDAR (10 Hz); 2xcolor cameras (10 Hz); 1GPS/IMU	L2C	2D-3DOD 2D-3DTK PD
SODA10M [21]	Multi	1080 P	—	2DOD
CU-LiDAR [22]	Car	2xVPL-16 (10 Hz); 2xSICK-LMS (100 Hz); 1xGPS (10 Hz); 1xFOG (1000 Hz); 1xIMU (100 Hz); 2xWheel; 1xAltimeter	ROS (s)	Lo
Malaga [23]	Car	1xStereo (20 Hz); 2xSICK-LMS (75 Hz); 3xHOKUYO (40 Hz); 1xGPS (1 Hz); 1xIMU (100 Hz)	G2S	Lo

Sync refers to synchronization, and ROS (s), C2C, L2C, and, G2S Indict the system time (ROS), camera-camera, LiDAR-camera, and GNSS triggering method. The abbreviation of tasks are following as localization (LO), object detection (OD), semantic segmentation (SS), instance segmentation (IS), point clouds segmentation (PP), depth estimation (DP), optical flow (FL), scene flow (association depth and optical flow, SC), disparity (stereo, St), end-to-end (imitation learning, E2E), tracking (TK), prediction (PD), segmentation with tracking (STK), lane detection (LD), drivable (DA), planning (PL).

stages are complete, data collection takes place along designated routes and within specified time, speed, and duration parameters. For datasets requiring annotation, additional labels are applied, with data quality maintained through iterative verification. Following this, the dataset undergoes rigorous correction and cleaning prior to release. Often, additional resources such as evaluation metrics, user guides, and documentation of model performance are provided to enhance usability and facilitate effective use within the research community.

III. REAL DATASET

The real-world data is collected by physical platforms in actual road scenarios, with no specific requirements regarding the equipment used by the collectors, the duration of data collection, or the quality of the data. This lack of standardization results in significant variations in the type, quantity, and quality of data collected by different entities. Consequently, the adaptability of tasks is limited by the characteristics of the data collectors; even when the same task is achievable, there can be discrepancies in data quantity, size, and even type among different datasets. Table I presents common real-world data for autonomous driving tasks, detailing the data collection platforms, sensor installation configurations, time syn-

chronization types, and the corresponding task adaptability information. In the platform attributes, we have endeavored to collect information regarding the number and types of vehicles. For instance, “20 Car” indicates 20 small cars, while “multi” signifies multiple vehicles, though their specific models remain unknown. The terms “min” and “mid” denote small and medium vehicle sizes, respectively, while “SUV” and “SPV” refer to vehicle categories without specifying particular brands. Datasets like KITTI and nuScenes provide information about specific vehicle models, which is highly beneficial for researchers interested in platform characteristics.

In Table I, we provide detailed information on the sensors used for data collection in each dataset, enabling readers to identify the types of data available. We include comprehensive details wherever possible, such as the number of sensors, their models, and capture frequencies. The “Sync” column indicates the time synchronization method, primarily relevant to GNSS, LiDAR, and cameras. Specifically, “L2C” denotes that the camera is triggered by LiDAR; “ROS (s)” implies that each sensor is triggered independently according to system frequency; “C2C” signifies synchronization only among cameras; and “G2S” represents GPS-based time-stamping for triggering each sensor. The “Task” column outlines the supported tasks, with abbreviations explained in the caption for

Table I.

Table II provides data volume information for various datasets. “Se” denotes the number of independent data streams within each dataset, while “D (p)” represents duration, typically as an average, given that each data stream’s length can vary. The last four columns indicate the total number of images, LiDAR point clouds, millimeter-wave radar point clouds, and GPS/IMU pose data included in each dataset. Based on Tables I and II, we can analyze that in major autonomous driving datasets, visual image data is the most prominent, as all datasets contain image information. With the deployment of LiDAR, these datasets can support tasks such as 3D detection and point cloud segmentation. However, 3D bounding boxes for objects can also be obtained without LiDAR data, through world perspective (roadside), direct acquisition from the target’s own pose estimation, or stereo cameras. Synchronizing cameras using LiDAR triggers is a common strategy in multi-sensor setups, and most datasets employ short-duration recording sessions with a wide variety of independent scenes. A key issue, however, is that differences in sensor types, data quality, and acquisition frequencies across datasets result in limited standardization and interoperability between them. In the remainder of this chapter, we will discuss commonly used data related to perception, decision-making, and control.

TABLE II
THE NUMBER OF FRAMES AND THE
TOTAL DATA VOLUME

Dataset	Se	D (p)	Im	Li	Ra	G/I
BDD100K [12]	100 K	40 s	120 M	–	–	120 M
D ² -City [13]	11 K	30 s	700 K	–	–	–
nuScenes [7]	1000	20 s	240 K	1.4 M	390 K	2 M
ONCE [15]	560	–	7 M	1 M	–	–
PandaSet [16]	103	8 s	50 K	16.5 K	–	–
Oxford [8]	–	6 m	20 M	–	–	–
CityScapes [18]	–	30 s	–	–	–	–
KITTI [5]	–	30 s	–	–	–	–
Waymo [6]	1150	20 s	1 M	230 K	920 K	–
Lyft-L5 [19]	350	60 m	47 M	110 M	–	–
SODA10M [21]	–	–	10 M	–	–	–
D ² -City [13]	10 K	30 s	7.5 M	–	–	–

Se, D (p), Im, Li, Ra, G/I refer to the number of scenes, the during time, the number of camera frames, LiDAR frames, reader frames, and GPS/IMU frames.

A. Perception

Perception is one of the most critical tasks in autonomous driving, with further sub-tasks that include localization, detection, segmentation, tracking, and prediction. Since perception data typically involves at least one sensor type, such as a camera or LiDAR, it often results in large volumes of data.

1) *Localization*: Depending on the availability of prior maps, localization can be divided into map-based localization methods and map-free localization strategies, with the latter being more based on simultaneous localization and mapping (SLAM) techniques in academic research. In map-based local-

ization methods, there are image geolocation and visual localization. Various datasets, including CVUSA [24], Aachen Day-Night [25], KITTI [5], UTBM [26], comma2k19 [27], and FORD [28], and etc., offer driving data and evaluation metrics for localization tasks.

For map-based localization, image geolocation is the process of determining the geographic location or coordinates (latitude and longitude) of an image based on its visual content. It involves matching the features or landmarks present in the image with a pre-existing map or geospatial database to estimate the image’s location. Charleston [29] is the first dataset on this task, which collects more than 6 K pairs around Charleston. San Francisco [30] (183 K), CVUSA [24] (43 K), CVACT [31] (92 K) and WITW (Google street view) are commonly used for image geolocation. Visual localization is another task belonging to the map-based strategy, which requires the visual data, maps created by 3D sensors, and outputs the 6-DoF posture of cameras. This task can be further divided into place recognition and visual localization. Place recognition involves matching a given image with images in a database to determine the initial location. Datasets suitable for place recognition include Nordland [32], Tokyo-24/7 [33] and Pittsburgh [34]. Visual localization dataset increases the 3D model of the environment for matching and poses truth for training and estimating such as Aachen [35], Michigan [36], and etc. Aachen Day-Night [25] extends the data in various weather conditions, night time, and different seasons.

For SLAM-based dataset as described with specific properties outlined in Table III with the information on collection year, institute, location, scene, the type of platform, running distance, range, during time, the number of scenes, and whether including visual, LiDAR data and supporting ranking. comma2k19 [27] is particularly suitable for evaluating visual-based tight-coupled localization and mapping algorithms. The Ford dataset [28] aims to advance the development of multi-agent systems localization and mapping. Oxford [8] and [26] offer the large-scale dataset encompassing over 1K km of odometry in Oxford’s city center and the downtown and suburban areas of Montbéliard, France, respectively, specifically designed for long-term localization tasks. KITTI [5], Waymo [6], BLVD [20] are popular datasets with multiple types of sensors in urban areas for localization. Queensland dataset [37] and Michigan dataset [36] collect data from a robot system on campus. It is important to note that visual localization tasks and visual SLAM tasks may have overlap in terms of datasets. This is because both tasks involve estimating a 6D pose. However, the main difference lies in the fact that visual localization heavily relies on 2D-3D conversion and map matching, does not prioritize real-time computational requirements, and has the capability to output the global location. Considering advancements in localization, it is crucial to prioritize datasets that cover relevant scenarios, including long-term SLAM [36], [38], dynamic SLAM [39], and underground SLAM [40], etc.

2) *Pedestrian, Lane, Road Marking, Traffic Signs and Drivable Region Detection*: These tasks focus on 2D object detection and segmentation, areas in which researchers are already demonstrating competitive results. The Caltech Pedestrians

TABLE III
THE DETAILED CHARACTERISTICS OF LOCALIZATION DATASET (SLAM-BASED)

Dataset	Year	Institute	State	Scene	Platform	Odom	Range	During	Scenes	Visual	LiDAR	Rank
comma2k19 [27]	2018	comma.ai	CA, US	Highway	Car	–	20 km	1 min	2019	√	√	√
Ford [28]	2020	Ford	MI, US	Multi	Multi	–	–	Multi	100+	√	√	√
Oxford [8]	2017	Oxford	Ox, UK	City	Car	–	1 km	6 min	–	√	√	√
UTBM [26]	2017	UTBM	Mont, Fr	Downtown	Car	5 km	63.4 km	–	11	√	√	√
KITTI [5]	2012	KIT	KL, GE	Multi	Car	–	39.2 km	–	22	√	√	√
CU-LiDAR [22]	2018	KAIST	Seou, Kr	City	Car	2 km	44 km	–	11	√	√	–
Malaga [23]	2013	Almeria	Cities, Sp	City	Car	–	36.8 km	4–40 min	5	–	√	–
Queensland [37]	2013	UQ	AUS	Campus	Robot	1.6 km	1.6 km	5 min	1	√	–	–
Michigan [36]	2012	UM	MI, US	Campus	Robot	–	147.4 km	–	27	√	√	–

dataset [4], released in 2009, comprises a 10-hour video with 250 K frames annotated with 350 K instance bounding boxes for pedestrians. This dataset pioneered the benchmark for pedestrian detection and highlighted the limitations of measurement methods at that time. The ETH Pedestrian dataset [41] (1894), TUD [42] (1.7 K), Daimler dataset [43] (72 K), Tsinghua-Daimler Cyclist data [44] (32 K), CityPersons [45] (35 K collects images from CityScapes [18]), NightOwls [46] (279 K at night), EuroCity [47] (238 K), KAIST Pedestrian [48] (103 K with night scenes) are optional datasets in this kind of tasks. The pedestrian detection task is currently advancing towards research focused on detecting and tracking pedestrians in increasingly diverse and complex environments.

The purpose of lane detection is to extract lane regions from images or videos, which can then be utilized in drivable region detection and semantic segmentation tasks. Recent research has focused on the hot topic of 3D line detection, which incorporates 3D information from LiDAR or radar. TuSimple provides approximately 7 K annotated frames. CurveLanes [49] comprises 150 K frames with curved and multiple lanes, collected in urban and highway environments in China. CULane [50] (133 K), VPGNet [51] (20 K with rainy and nighttime scenarios) are other datasets for lane detection. LLAMAS [52] (100 K), OpenLane [53], and ONCE-3DLanes [54] are suitable for 3D lane detection. For more detailed information, please refer to Table IV.

The road marking dataset [59] serves as the pioneering dataset for road mark detection and recognition tasks. It includes 29 individual videos captured in the US, comprising a total of 28K frames with 1208 annotated 2D regions. The following are other related datasets including TRoM [60] (712 in 19 categories), BDD100K [12] (100 K in 8 categories), CeyMo [61] (4706 labels in 2887 frames with 11 categories). Traffic sign detection is a task that focuses on classifying cropped sign figures or detecting signs from raw images. RSD dataset consists of 877 cropped sign images across 4 categories, making it suitable for sign classification methods. Some common data for this tasks are listed: BSTLD [62] dataset (10.7 K signs in 5 K images), DFG Dataset [63] (13 K in 7 K with 200 categories), Chinese traffic sign database (CTSD), Mapillary traffic sign dataset (MTSD) [64] (100 K images with 300 categories), GTSD [65] (1 K images in

TABLE IV
THE DETAILED CHARACTERISTICS OF LANE DETECTION DATASET

Dataset	Fra	Seq	We	Ni	3D	Scs	Ra
TuSimple	7 K	7 K	–	–	–	HW	√
CurveLanes [49]	150 K	–	√	√	–	Multi	√
CULane [50]	133 K	–	√	√	–	Multi	√
VPGNet [51]	133 K	4	√	√	–	UB	–
LLAMAS [52]	100 K	–	√	√	√	Multi	√
BDD100K [12]	10 K	10 K	√	√	–	Multi	√
Caltech [4]	1225	4	–	–	–	UB	–
JiqingHW	210 K	40	–	√	–	HW	–
LaneISUE [55]	24 K	402	–	–	–	Multi	–
DET [56]	5 K	–	–	–	–	UB	–
Comma-LD [57]	2 K	100	–	√	–	HW	–
SDLane [58]	43 K	–	–	√	–	Multi	√
OpenLane [53]	200 K	1 K	√	√	√	Multi	√
ONCE-3DLanes [54]	211 K	–	√	√	√	Multi	√

Fra, Seq, We, Ni, 3D, Scs, Ra refer to the number of frames, and individual sequences, whether including various weather, night data, 3D information, the scene location, and whether providing ranking. HW and UB represents highway and urban areas.

Germany), CCTSDB [66] (16 K images across 3 categories in China), TSD [67] (3 K in 2 K with 6 categories in Linköping), Tsinghua-Tencent [68] (100 K images). Several datasets focus on traffic light detection, such as KUL [69], German traffic sign [70], highway workzones [71], Bosch [62], LaRA [72], and DriveU traffic light [73] (230 K boxes in 188 videos).

Drivable region detection is a crucial aspect of driving tasks and can be considered a specific segmentation task. BDD100K [12] dataset provides annotations for drivable pixels in 100 K frames, classifying them into two categories: directly drivable areas and alternative areas. However, it is important to note that the number of publicly available datasets applicable to drivable area recognition is limited. From an academic standpoint, static detection tasks such as drivable areas, lane lines, and road sign recognition are often integrated into the pavement segmentation task. It is worth mentioning that these tasks are critical for the implementation of commercial functions at the L2 level of autonomous driving vehicles.

TABLE V
THE STATISTICS OF THE 2D SEGMENTATION AUTONOMOUS DRIVING DATASET

Dataset	Place	Cat	Wea	Day	Frame	Cam	Scs	Resolution	Sem	Ins	3D	Extra
CamVid [75]	Camb, UK	32	–	–	701	1	5	960 × 720	√	–	–	3D pose
CityScapes [18]	cities, DE	7	–	–	5 K	2	–	–	√	√	–	3D pose
Mapillary [77]	World	66 (37)	√	D/N	25 K	Multi	25 K	>1920 × 1080	√	√	–	–
BDD100K [12]	cities, US	40	√	D/N	10 K	–	10 K	720p	√	√	–	3D pose
Apollo [17]	BJ, CHN	25	√	D/N	144 K	2	1	3384 × 2710	√	√	√	3D po/Dp
IDD [82]	areas, IN	34	–	D	10 K	1	182	1678 × 968	√	√	–	–
SemanticKITTI [84]	Kr, DE	25	√	D	43 K	–	22	–	–	–	√	–
Highway Driving [78]	Seoul, KR	10	–	D	1.2 K	1	20	–	√	–	–	–
DriveSeg [76]	–	19	–	D	5 K	1	1	1920 × 1080	√	–	–	30FPS
A2D2 [10]	South, DE	38	√	D	41.2 K	6	–	1928 × 1208	√	√	√	3D box
Waymo [6]	cities, US	28	√	D/N	100 K	–	2 K	1920 × 1080	√	√	√	3D po/box
PandaSet [16]	SiVa, US	37	√	D/N	6 K	6	76	–	–	–	√	3D po/box

Cat: categories, Wea: weather, Cam: camera frames, Scs: the number of different scenes, Sem: semantic segmentation, Ins: instance segmentation, 3D: point cloud segmentation. Extra means that some of data provide extra data such as 3D pose of the ego-vehicle (3D po), depth maps (Dp), 3D bounding boxes of targets (3D box).

3) *Scene Classification and Understanding*: Scene classification in autonomous driving involves categorizing the perceived scene into sub-categories based on factors such as location, daytime, weather, congestion conditions, etc. And the CompCars [74] with 1716 car models in 136 K images 27 K cropped images, and BDD100K [12] dataset with 100 K images in 6 weather conditions in 6 scene categories are common used in this task.

4) *2D & 3D Detection and Segmentation*: 2D semantic or instance segmentation in autonomous driving refers to the task of classifying each pixel in an image into a specific category or assigning it an individual identification number. 3D segmentation involves performing similar tasks on point cloud data obtained from LiDAR sensors. CamVid [75] is the first dataset for semantic segmentation in autonomous driving, which collects over 10 min of monocular videos at 30 Hz frequency, and selects key frames in 1 Hz or 15 Hz frequency as segmentation images. A2D2 [10] provides 41.2 K annotated frames from 6 different cameras installed on the top of the collection platform, which is suitable for multiple view tasks. It also offers 12.5 K annotated 3D boxes for association tasks and 38.4 K 3D points labels for point cloud segmentation. DriveSeg [76], Mapillary [77] (66 categories in semantic and 37 categories instance segmentation across multiple platforms), Cityscapes [18] (5 K images with fine and 20 K images with rough annotations), Highway Driving [78] (20 2 s videos in 30 Hz), Raincouver [79], [80], and Rainy WCity [81] (rainy weather), IDD [82] (unstructured roads), Wild-dash [83] (1800 frames) are other datasets for segmentation.

Point cloud segmentation refers to assigning class labels to each point in a point cloud. SemanticKITTI [84] is a dataset that builds upon the vision odometry benchmark of KITTI [5] and provides annotations for each point in sequential frames, enabling point cloud segmentation. Besides SemanticKITTI, PandaSet [16], Apollo [17] are widely used for the development in point cloud segmentation. The detailed information on segmentation datasets can be seen in Table V.

BDD100K [12] is a dataset suitable for 2D object detection,

providing 100 K annotated frames with 10 categories. D^2 -City [13] (700 K frames from 1 K 30 s videos with 12 categories from 500 different platforms in 5 cities in China), JAAD [85] (378 K bounding boxes in 3 K instances from 346 10 s videos), TME Motorway [86] (28 1 min videos on highways), [87] (600 K annotations from 200 K images), Boxy [88] (1.99 M annotated labels in 200 K images), SODA10M [21] (20 K images from 32 cities by phones in China), FLIR (520 K annotation in 26 K with thermal camera data), multi-spectral data [89] (with thermal camera data) are other data for 2D object detection.

KITTI [5] is a pioneering benchmark dataset for evaluating algorithms for 3D object detection, collected by the Karlsruhe Institute of Technology (KIT) in 2011 with 80 K annotations among 15 K frames. A*3D [9] provides more complex scenarios for 3D object detection, including nearly one-third of the data captured at night, 17% high-density data, and a significant portion of occluded targets (25%). A2D2 [10], H3D [14], nuScenes [7] (360-view), ONCE [15] (1 M frames from 560 videos), PandaSet [16] (8 K frames in 103 videos with a mechanical spinning and a forward-facing LiDAR), Waymo [6] are primarily used for 3D tasks. The detailed information on 3D object detection can be seen in Table VI. Based on the collected information, we observe that segmentation and detection datasets are evolving toward higher data volumes and greater diversity, encompassing multiple climates, varied environments, different platforms, and a range of tasks.

Furthermore, it is worth noting that 3D bounding boxes in 3D detection cannot accurately represent object contours. As a result, researchers have placed emphasis on voxel-based occupancy methods, which have benefited from the advancements in surround-view vision, birds' eye view (BEV) techniques, and Transformer architectures. However, the sparsity and disorderliness of point cloud data pose challenges for the subsequent development of this task. In response, OCC-based datasets have emerged, including OCC3D [90], OpenOccupancy [91], etc. The metrics of 3D occupancy are similar to the point segmentation tasks.

TABLE VI
THE STATISTICS OF THE 3D OBJECT DETECTION AUTONOMOUS DRIVING DATASET

Dataset	Year	Institute	Place	Classes	Weather	Day	Frame	Boxes	Rank	FoV	Seq	DoF	Range
KITTI [5]	2012	KIT	Germany	9	–	D	15 K	80 K	√	90	√	7	70 m
A*3D [9]	2020	A*Star	Singapore	7	√	D/N	39 K	230 K	–	57.3	–	7	100 m
A2D2 [10]	2020	Audi	Germany	38	√	D		12.5 K	–	60	–	7	100 m
H3D [14]	2019	HondaRI	Cali, US	8	–	D	27.7 K	1 M	–	360	√	7	100 m
nuScenes [7]	2019	Motinal	Bos, US&SGP	23	√	D/N	40 K	1.4 M	–	360	√	7	80 m
ONCE [15]	2021	CUHK	cities, CHN	5	√	D/N	16 K	417 K	√	360	√	7	200 m
PandaSet [16]	2021	Hesai	SiVa, US	28	√	D/N	8 K	–	√	360	√	7	300 m
Waymo [6]	2020	Waymo	Cities, US	4	√	D/N	390 K	12.6 M	√	125	√	7	75 m

Seq refers to whether providing the sequential frames. FoV, Seq, DoF, and Range refer the field of views on cameras, whether having sequential data, the target's degrees of freedom and valid range for detection system.

5) *Disparity, Flow Estimation, and Depth Estimation*: disparity estimation, also known as binocular estimation, is a crucial aspect of 3D computer vision. It involves calculating the offset of the same pixel in the corrected left and right images. By utilizing a known binocular baseline, the depth map can be obtained through calculations. KITTI [5] is the first dataset to provide a task for binocular parallax estimation in unmanned scenarios. The dataset includes 194 images specifically selected for binocular and optical flow tests. DrivingStereo [92] consists of 42 sequences with 10 M frames and introduces a novel metric that is more suitable for driving scenarios, enhancing the evaluation of binocular estimation methods. Stereo-Event [93] (including 2 event cameras, 2 color cameras, a LiDAR, and an IMU to cover various 3D and visual tasks), CVC-13 [94] (2 Sony ICX084 Bayer cameras and an infrared camera), CCSAD [95] (96 K frames with IMU and GPS data) are other data commonly used in this task. Optical flow estimation aims to determine the motion of objects in a sequence of images or video frames. KITTI [5] provides 389 images and consecutive frames specifically for optical flow estimation. KITTI-2015 [96] expanded the previous stereo/optical dataset by incorporating motion targets. HCI [97] released 55 19 to 100 frames stereo pairs with disparity and optical flow information. Depth estimation estimates the distance of pixels in an image and KITTI-Depth [98] divides the task into depth completion and depth estimation with 94 K dense depth maps, generated by fusing the results of laser scans and stereo reconstruction using a technique called semi-global matching (SGM) [99]. These tasks, which provide complete scene information instead of direct target-specific outputs, are often less suited for outdoor autonomous driving applications, where targeted data is generally more useful. As a result, many algorithms in this domain remain predominantly confined to indoor research settings.

6) *Tracking and Prediction*: Argoverse [11] is widely used in 3D tracking and prediction tasks. This dataset provides an HD map covering 300 km of odometry in Pittsburgh and Miami, including lane, drivable area, and rich semantic information of the road. BDD100K [12], nuScenes [7], Waymo, and METEOR [100] are commonly used in tracking tasks (Table VII). Besides, multi-object tracking and segmentation (MOTS) is a novel task to eliminate the ambiguities from occluded bounding boxes. MOTS [101] combines the track-

ing and segmentation results on KITTI [5] and releases a novel dataset on MOTS with the semi-automated method. BDD100K [12] provides richer data with 14 K frames, and DGL-MOTS [102] has advantages in annotation quality and scene diversity (Table VIII). For Prediction, in Argoverse [11], more than 324 K 5 s sequences in 10 Hz are selected. Lyft-L5 [19] includes 170 K 25 s videos with various attributes (BEV boxes, velocity, acceleration, angle velocity, category). Besides, it also provides an HD map with 15.2 K annotated road segments and an aerial map in 10560×10560 , covering 74 km². BLVD [20] releases 249 K 3D labels and 5 K independent IDs, and TRAF [103] includes 50 visual sequences for prediction in dense and heterogeneous scenarios. Datasets that integrate perception, prediction, and tracking are receiving increasing attention.

TABLE VII
THE DETAILED INFORMATION OF TRACKING DATASETS

Dataset	Fr	Se	D (p)	ID	Box	2D	3D
Argoverse [11]	20 K	113	~20 s	11 K	–	√	√
BDD100K [12]	318 K	2 K	40 s	130.6 K	3.3 M	√	–
D ² -City [13]	700 K	1 K	30 s	–	–	√	–
H3D [14]	27.7 K	160	20 s	–	1.4 M	√	√
nuScenes [7]	40 K	1 K	20 s	–	1 M	√	√
Waymo [6]	390 K	2 K	20 s	–	12.6 M	√	√
BLVD [20]	–	654	20 s	5 K	214 K	√	√
METEOR [100]	2 M	1250	1 min	5 K	214 K	√	√

Fr: the number of frames, Se: the number of individual sequences, D (p): during the time of each video. 2D and 3D indict whether the dataset provides 2D and 3D tracking.

B. Planning & End2End & Imitation Learning

Planning and decision-making are downstream tasks of perception, which generally require knowledge of the vehicle's position, target state, and destination information. Since real-world data does not allow for online control, decision-making tasks are often approached through end-to-end or imitation learning frameworks. Vehicle attributes for these tasks are typically obtained via CAN-BUS, dashboard data, or pose estimation methods.

Lyft-L5 [19] uses real collected data to learn self-vehicle path planning. In the 25 s video, the self-vehicle trajectory

TABLE VIII
THE DETAILED INFORMATION OF MOTS DATASETS

Dataset	Fr	Se	D (p)	ID	Mask
MOTS [101]	8 K	21	–	749	38 K
BDD100K [12]	14 K	70	40 s	6.3 K	129 K
DGL-MOTS [102]	12 K	40	20 s	1.6 K	68 K

path is displayed on the BEV diagram. nuPlan [104] is the first large-scale planning dataset for open-loop and closed-loop methods with 1.3 K driving hours at 4 cities in the US. The planning evaluation without vehicle control (open-loop) could be adopted in the recorded bags, and planning with control could be evaluated in the simulator [105]. Brain4Cars [106] consists of 700 events with 3.5 s of videos. DIPLECS [107] is a simple dataset for end-to-end task with 2 30 min videos (2 cameras, a steering wheel recorder, and an odometry recorder). DBNet [108] (10 K frames in 300 videos with a LiDAR, a camera, and steering with speed data), DDD17 [109], UAH-DriveSet [110] (500 min with front images, GPS, IMU, and maneuvers data from a smartphone application, DriveSafe), HDD [111] (137 45 min frames), StreetLearn [112] are the data suitable for the above task. The raw data of BDD100K [12] (100 K 40 s sequences in 30 Hz from cameras and GPS/IMU) can be utilized to develop the imitation learning methods. Through statistical analysis, we found that data supporting the above tasks is significantly less abundant than perception-related data. Achieving these tasks often requires low-level vehicle data, yet vehicle control attributes determined through pose estimation exhibit considerable inaccuracies. Consequently, downstream tasks for vehicles are more frequently conducted on simulation platforms with synthetic data.

C. Vehicle-to-Anything (V2X)

Autonomous driving safety is faced with great challenges. Problems such as blind driving areas and instability of middle and long distance perception exist in single-vehicle intelligence, which leads to limited operating design domain (ODD) of autonomous vehicles and blocked landing of single-vehicle intelligent autonomous driving. Cooperation between vehicles and roads will be the only way to ensure the safe operation of autonomous driving. Tsinghua University and Baidu release a series of datasets on V2X, including Rope3D [113] (5 K images with 3D labels from roadside) and V2X-Seq [114]. The latter data is the first time-series perception dataset of vehicle-road coordination in real scenarios, including 100 20 s videos with point clouds and images in 10 Hz from both the vehicle side and the roadside platforms. V2X data is also among the most popular data types today. Extensive collection of diverse scenarios and information sources can significantly contribute to the advancement of intelligent transportation systems.

D. Drivers Action Detection & Intelligent Transportation Analysis

Driver's action detection refers to the process of identifying and analyzing the actions or behaviors of a driver while operating a vehicle, often using sensors, cameras, or other moni-

toring devices. The DrivFace contains 600 images in 4 individual sequences in real scenarios with 2D boxes, facial key points, and eye direction. Intelligent traffic analysis is primarily used to evaluate the status of the transportation system from the perspective of surveillance or aircraft. NYC3DCars [115] collects thousands of figures from multiple surveillance cameras at three different cities in the US with geospatial and target 2D, 3D location, and orientation information. VeRi-776 [116] is a re-identification dataset that comprises 49 357 images of 776 vehicles captured from 20 cameras. VERI-Wild (40 K IDs in 400 K frames from 174 cameras) [117], CityFlow [118] (666 IDs from 40 cameras), and VehicleID (26 K IDs in 200 K frames from 2 cameras) [119] are also suitable for the re-identification task. BoxCars116k [120] collects 27.5 K targets from 116 K frames selected by 137 surveillance cameras with 693 categories, pseudo-3D bounding boxes, and foreground mask information in Brno, Czech. UA-DETRAC [121] dataset releases 100 videos (140 K frames, 8 K IDs, and 1.21 M labels) from the roadside camera in 25 Hz. The highD [122], NGSIM [123], and INTERACTION [124] are the datasets from the BEV of surveillance or drones for multiple vehicle agents prediction, ego-vehicle path planning, and imitation learning [125]. The difference between intelligent transportation analysis data and V2X lies in that the former does not focus on improving the accuracy of the vehicle's perception or control results, but rather on global quantitative analysis.

E. Large Language Models Dataset in Autonomous Driving

Several notable datasets have contributed to advancing large language models (LLMs) and interpretability research in autonomous driving (with language description). Refer-KITTI [126], leveraging the KITTI [5] dataset, introduced language expressions as semantic cues to guide multi-object tracking prediction models, thereby contributing to the progression of multi-object tracking algorithms. Cityscapes-Ref [127] introduced additional annotations related to language expressions across entire video sequences on the original CityScapes [18] dataset, thereby enriching perceptual information. nuScenes, widely employed for multimodal tasks due to its extensive surround-view data, extended its utility to the domain of autonomous driving with nuScenes-QA [128], thereby establishing a foundation for research on human-machine interaction in autonomous driving. Talk2Car [129] represents an end-to-end task describing the comprehensive behavior of the ego vehicle but providing descriptions for only a single target. NuPrompt [130] enhances the descriptive content by offering language explanations for multi-object tracking. BDD-X [131] and BDD-OIA [132] provide behavior-based explanations based on BDD-100K [12], enabling the analysis of reasons for vehicle behavior. HAD [133] offers natural language suggestions for vehicles, lacking specific vehicle behaviors but guiding control models through visual attention mechanisms to anticipate potentially expected behaviors. NuInstruct [134] and OpenDV-2K [135] are suitable for video QA and description in driving scenarios. Expanding existing datasets by adding textual descriptions is a strategy that revitalizes old data.

F. Other Interesting Data

Road damage dataset [136] collects 26 K images to detect the damaged regions on the road in 2D boxes such as cracks, and potholes. Lost and found [137] adopts a ZED stereo camera to capture 112 videos and picks 2104 images to annotate the pixel-level attribute belonging to 37 small obstacles. RoSA [138] introduces a construct zone segmentation data. CDNet [139] introduces a novel network for crosswalk detection and releases special data with 6868 annotated crosswalks in 2D boxes. CLPD is a dataset for car license detection, which includes 433 cropped vehicle images with the license 2D bounding box label in VOC format. CRPD [140] focuses on the license detection task in Chinese road scenarios. Drive attention prediction aims to detect the weight value of each pixel for the current driving scenarios and the purpose of this task is to detect the hidden risks and provide guidance for the following task including DR (eye)VE [141] (74 5 min videos with 555 K frames, BDD-A [142]. DADA-2000 [143] selects 2 K individual videos to achieve the task in accidental situations recognition. DeepScenario [144] and ROAD [145] are released for failure detection and analysis. These datasets address the edge cases in autonomous driving from different perspectives. The typical tasks and associated datasets can be seen in Fig. 4.

Localization comma2k19/Ford/ Oxford/KITTI	Traffic Sign RSD/CTSD/ /KUL/DriveU	Point Seg SemanticKITTI/ PandaSet/Apollo	Optical KITTI-2015/ HCI	MOTs MOTS/ BDD-100K	Imitation Brain4Cars/DDD17 /HDD/BDD100K
Pedestrian Det Caltech/ETH/ TUD/Daimler	Drivable BDD-100K	2D Detection KITTI/ID-City/ JAAD/SODA10M	Depth KITTI	Driver DriveFace	V2X Rope3D/V2X-Seq
Lane Det TuSimple/CULane/ OpenLane	Understanding CompCars/ BDD-100K	3D Detection nuScenes/Waymo A2D2/H3D	Tracking Argoverse /nuScenes	Traffic NYC3DCars/ VeRi-776/VehicleID	License CLPD/CRPD
Road Mark Road Marking /TroM/BDD-100K	Segmentation CityScape/ Mapillary/DriveSeg	Stereo KITTI/ DrivingStereo	Prediction Argoverse /Lyft-L5	Planning Lyft-L5 /nuScenes	Novel DADA-2000/ DeepScenario

Fig. 4. This figure presents an overview of the driving tasks along with the associated datasets.

IV. SIMULATION PLATFORMS

Simulation platforms in autonomous driving allow researchers to develop the environment by themselves. These simulators can be divided into vehicle and traffic based platforms.

A. Vehicle-Based Simulator

Vehicle-based simulator are designed to develop and evaluate the performance of downstream algorithms such as path planning and control. they can be further divided into two types: 1) engine-driven and 2) data-driven. The engine-driven platforms such as CARLA [146] allow users to construct their arbitrary driving environments and regulations, and assist users in locating model problems, however, require them to build the entire system and are difficult to deploy in real scenarios. Data-driven platforms such as Apollo [17], and Auto-ware, collect data from realistic scenarios, but the testing capabilities are limited and hidden problems are hard to recognize. In addition, there are four different test methods: Model in the loop (MIL), software in the loop (SIL), processor in the loop (PIL), and hardware in the loop (HIL) [147].

B. Traffic-Based Simulator

Traffic simulation is the application of computer simulation in traffic engineering. Its purpose is to establish a virtual reality traffic computer model, on the basis of the model using computer technology to reproduce complex traffic phenomena, and explain and analyze these phenomena, find out the crux of the problem, so as to evaluate the pros and cons of traffic control scheme and the rationality of traffic planning and design. It is a crucial loop for intelligent traffic and smart cities, such as PTV Vissim, TESS, VTD, and SUMO [148]. The detailed information can be seen in Table IX.

V. VIRTUAL DATASET

Virtual data is acquired from acquisition equipment built on a simulation platform and collected according to established rules, as shown in Table X. Its purpose is to compensate for the lack of real data by providing additional data. Its advantages include full automation of annotation, high-precision synchronization, and rich data scenes and modes. However, it introduces differences compared to the real scene, which requires researchers to pay attention to the transformation from virtual to real.

A few virtual datasets are created for lane detection or 3D lane detection, including CARLANE [162], Gen-LaneNet [163] and 3D-LaneNet [164]. CURE-TSR [165] collects 17 M images with more than 20M traffic signs. SYNTHIA [154] is a virtual dataset collected from Unity for semantic segmentation with 13 categories. Virtual KITTI [155] involves 50 individual videos with 21 K frames generated from KITTI by the real-to-virtual cloning method for 2D and 3D multi-object tracking, semantic, instance segmentation with 15 categories, flow, and depth estimation tasks. PfBench [156] collects in GTA-5 with 250 K continuous video frames for 3D object detection and tracking, segmentation, etc. LIVOX pays more attention to LIVOX LiDAR point cloud data with full views in 14 categories. OPV2V [167] is a LiDAR-based dataset with 233 K annotated instances in 11 K frames from multiple platforms for the V2V detection task.

VI. EVALUATION METRICS

In this chapter, we categorize metrics into general metrics and driving-driven metrics (as shown in Fig. 5). The former aligns with the metrics used in computer vision and robotics, while the latter is specific to or primarily used in the autonomous driving domain.

A. General Metrics

The Top@K metric [168] is commonly used in image geolocalization, place recognition, scene classification, and understanding tasks to evaluate the performance of algorithms. Following [25], [169], position and orientation errors are determined by the Euclidean distance and rotation deviation from the ground truth; these metrics are widely used to evaluate visual localization and SLAM performance. Absolute trajectory error (ATE) introduced in [5] is also used in the SLAM evaluation process. The 2D object detection tasks (including pedestrian, traffic signs detection, etc.) adopt Preci-

TABLE IX
TYPICAL SIMULATION PLATFORMS

Name	Ins	Model	M-A	Sces	Sens	3DV	WT	NT	Dy	V-R	V-X	Base	Traf	Func
CARLA [146]	AUB	Engine	√	√	√	√	√	√	√	√	√	Unreal4	–	–
PreScan	Siemens	Engine	√	√	√	√	√	√	√	√	√	Simulink	–	S H
AirSim [149]	MicS	Engine	√	√	√	√	√	√	√	√	–	Unreal4	–	S H
Udacity [150]	Uda	Engine	–	√	√	√	√	√	√	–	–	Unity	–	M S
LGSVL [151]	LG	Engine	√	√	√	√	√	√	√	√	√	Unity	–	S H
Apollo [17]	Baidu	Data	–	√	√	√	√	√	√	–	–	Unity	–	M S
CarMaker	IPG	Engine	–	√	–	√	–	–	√	√	–	Unity	–	H
PTV Vissim	PTV	Engine	–	√	–	√	–	–	√	–	–	–	√	–
TESS	Tongji U	Engine	–	√	–	√	–	–	√	–	–	–	√	–
Panosim	–	Engine	–	√	√	√	√	√	√	√	√	–	–	S H
VTD	VIRES	Engine	√	√	√	√	√	√	√	√	–	–	√	S H
DRIVE-C	NVIDIA	Engine	–	√	√	√	√	√	√	√	–	–	–	S H
CarSim	MS	Engine	–	√	–	√	–	–	√	√	–	Simulink	–	H
51Sim-One	51VR	Engine	√	√	√	√	√	√	√	√	√	–	–	M S
Parallel	Domain	Engine	–	√	–	√	√	√	–	–	–	–	–	–
RightHook	RH	Engine	–	√	√	√	√	√	√	√	–	–	–	S H
SUMO [148]	GNSC	Engine	–	√	–	–	–	–	–	–	–	–	√	–
MetaDrive [152]	HKSAR	Engine	–	√	√	√	√	√	√	–	√	Gym	–	M S
Traffic3D [153]	Aston	Engine	√	√	√	√	√	√	√	√	–	Unity	–	M S

Ins: institute, M-A: multiple agents, Sces: multi-scene, Sens: multi-sensor, 3DV: 3D virtualization, WT: weather, NT: night, Dy: dynamic models, V-R: virtual reality connection, Traf: traffic flow estimation, Func: testing function.

TABLE X
THE SYSTEM, TASKS, AND RELATIVE INFORMATION ABOUT VIRTUAL DATASETS

Dataset	Sys	Frame	Sensor	Task
SYNTHIA [154]	Unity	213 K	8xCam	SS
Virtual KITTI [155]	KITTI	21 K	1xCam	2D-3DTK SS IS Fl Dp
PfBench [156]	GTA-5	250 K	1xCam	SS IS 2D-3DOD 2D-3DTK Dp Fl Sc Lo
AIODrive [157]	Carla	100 K	Full	2D-3DOD 2D-3DTK PD SS IS PP Dp PPD
GTA-V [158]	GTA-5	25 K	1xCam	SS IS
PreSIL [159]	GTA-5	50 K	1xCam; 1xLiD	2D-3DOD SS IS PP
Sim-V2X [160]	CARLA	20 K	6xCam; 1xLiD	2D-3DOD 2D-3DTK Dp SS IS PP
IDDA [161]	CARLA	1 M	1xCam	SS IS Dp
CARLANE [162]	CARLA	118 K	1xCam	LD
Gen-LaneNet [163]	Unity	6 K	1xCam	3DLD
3D-LaneNet [164]	blender	306 K	1xCam	3DLD
CURE-TSR [165]	–	17 M	1xCam	TS
SHIFT [166]	CARLA	2.5 M	5xCam; 1xSte; 1xOpt; 1xDep; 1xGPS; 1xIMU	2D-3DOD 2D-3DTK SS IS Fl Dp PP Lo PD PE
LIVOX	–	14 K	5xHori; 1xTele	3DTK PP
OPV2V [167]	CARLA	11 K	4xCam; 1xLiD; 1xGPS; 1xIMU	3DOD

sion, Recall, and F1 score, AP, mAP in Pascal-VOC [170] and COCO [171] by deciding the TP, FN, etc. For 2D semantic segmentation, the IoU score could be adopted to estimate the semantic segmentation model. CityScapes [18] extends the IoU metrics into instance IoU, $iIoU = iTP/(iTP + FP + iFN)$ with category i . Disparity, flow, and depth estimation metrics are similar to the methods in segmentation. KITTI [5] counts errors when the disparity exceeds 3 pixels and 5% of its true

value and calculates the percentage of erroneous pixels in non-occluded areas and the total for the disparity and flow. DrivingStereo [92] introduces a new distance-aware and a semantic-aware metric. KITTI-2015 [96] updates the evaluate metrics by considering the back-mapping but it is still general metric and can be used in other scenarios. The metrics of depth estimation come from [172], which could be used to evaluate both outdoor and indoor scenes with mean squared

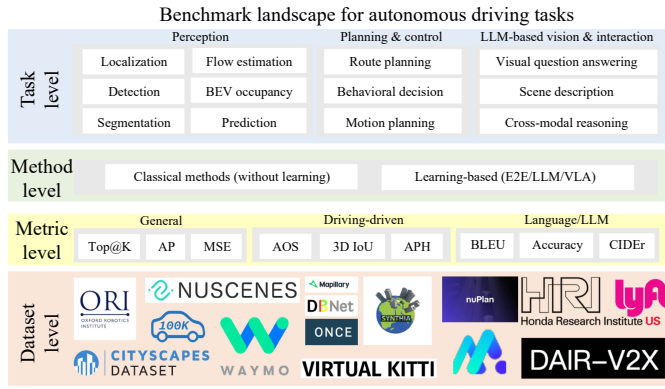


Fig. 5. Taxonomy of tasks, methods, evaluation metrics, and datasets in autonomous driving.

error (MSE) and mean absolute error (MAE), inverse root mean squared error (iRMSE), scale invariant logarithmic error (SILog), etc. The metrics for MOT are still from general vision, such as multiple object tracking accuracy (MOTA) [173], [174]. IDF1, IDFN, IDFP, and HOTA [175] are also calculated for tracking in a comprehensive way.

The metrics on prediction and planning tasks learn from robotics. Providing a sequence of previous states in BEV $V^t = (X^t, Y^t), t \in 0, \dots, T_{\text{cur}}$ and predicting the following states $F^t = (X^t, Y^t), t \in T_{\text{cur}} + 1, \dots, T_{\text{end}}$, the average displacement error (ADE), and miss rate (MR) can be calculated following [11], [20], [176]. The planning method only outputs a long trajectory from the initial point to the destination of the ego vehicle. Time-to-collision (TTC), time exposed time-to-collision (TET), and post encroachment time (PET) are common used following [104].

B. Driving-Driven Metrics

Lane detection and segmentation are similar tasks, but using computer vision methods makes it difficult to effectively evaluate detection performance, as they do not consider the full image context. CULane [50] sets the width in 30 pixels for each lane and computes IoU at pixel level to distinguish the positives and negatives. To solve the unfriendly downstream missions and consistent in gravity problems by adopting the Accuracy and F1, Comma-LD [57] introduces an end-to-end lateral deviation metric (E2E-LD) $\max_{t \leq T_E} (|L_t - C_t|)$ and a simplified E2E-LD metric, per-frame simulated lateral deviation metric (PSLD) $\frac{1}{T_P} \max_{1 \leq t \leq T_P} (|\tilde{L}_t - C_t|)$, where L_t is the lateral coordinate of the vehicle at time t , $\tilde{L}_t$ is the simulated lateral coordinate of the vehicle, and C_t is the lane center lateral coordinate corresponding to the vehicle position at time t . T_E, T_P refer to the durations. The metrics consider the vehicle's pose and the control methods for automated lane centering (ALC), a level-2 function for autonomous driving.

For the 3D lane detection metric, 3D-LaneNet [164] extends the 2D lane detection in the 3D space, which judges the positive/negative samples by Euclidean distance of the curve. Gen-LaneNet [163] provides a square root pointwise distance method as $\sqrt{\sum_i^n d_i^{jk}}$, where

$$d_i^{jk} = \begin{cases} (x_i^j - x_i^k)^2 + (z_i^j - z_i^k)^2, & \text{if } v_i^j = 1 \text{ and } v_i^k = 1 \\ 0, & \text{if } v_i^j = 0 \text{ and } v_i^k = 0 \\ d_{\max}, & \text{otherwise} \end{cases}$$

where v_i indicates whether the y -position (from 0 to n) of the image is matched to a truth. x_i, z_i are the position values of the x - z coordinate system in 3D space. $d_{\max}$ is a hyper-parameter during the experiments.

For orientation validation in 3D object detection, KITTI [5] designs AOS metric to evaluate the orientation estimation performance

$$AOS = \frac{1}{11} \sum_{r \in \{0.0, 0.1, \dots, 1\}} \max_{\tilde{r}: \tilde{r} \geq r} s(\tilde{r}) \quad (1)$$

where $r = TP/(TP + FN)$. $\mathcal{D}(r)$ denotes the set of predicts at recall rate r and $\Delta_{\theta}^{(i)}$ represents the difference between the matched i th orientation estimated and the truth.

$$s(r) = \frac{1}{|\mathcal{D}(r)|} \sum_{i \in \mathcal{D}(r)} \frac{1 + \cos \Delta_{\theta}^{(i)}}{2} \delta_i. \quad (2)$$

The evaluation process starts with calculating the BEV IoU to derive AP_{BEV} ; the final AP_{3D} is then calculated by extending the BEV match with the vertical intersection height between predicted and ground-truth objects. Influenced by [177], KITTI has adopted a strategy of using 41 discrete points $\{1/40, 2/40, \dots, 1\}$ to approximate the area under the curve for the evaluation of AP and AP_{3D} , instead of only the 11 sampled points. This modification eliminates the issue of fake precision encountered at the lowest recall bin $\{0, 0.1\}$, thereby enhancing the accuracy of AP calculation. nuScenes [7] introduce a novel metric for 3D object detection, which is more friendly for visual-based 3D object detection methods. It adopts the 2D center distance between the predictions and truth d on the 3D space instead of IoU. ONCE [15] proposes a novel calculation method for orientation evaluation, aiming to solve the problem that the 3D IoU formula could not deal with the adverse orientation in KITTI. Based on the IoU filter, the providers require that if the orientation deviation between the matched prediction and the truth is greater than 90 degrees, the prediction will be reassigned as negative samples, and the AP by P-R curve $p(r)$ as

$$AP_{3D}^{\text{Ori}} = 100 \int_0^1 \max \{p(r') \mid r' \geq r\} dr \quad (3)$$

where r represents the recall from 0.02 to 1 in the 0.02 step. Waymo [6] also detects problems with KITTI's judgment on orientation, and introduces a novel metric the accurate prediction on heading (APH):

$$APH = 100 \int_0^1 \max \{h(r') \mid r' \geq r\} dr \quad (4)$$

where the providers draw a P-R curve of orientation ($h(r)$), where positive and negative samples are judged based on $(|\hat{\theta} - \theta|, 2\pi - |\hat{\theta} - \theta|)/\pi$, the $\hat{\theta}, \theta$ represent the predicted and truth orientations. For tracking, Argoverse [11] adopts the center distance of the predicted and ground truth, orientation similarity, and amodal shape discrimination by 3D-IoU [178] to esti-

mate the tracking methods, which are more suitable for driving scenes. nuScenes [7] updates the MOTA inspired by [179] and introduces detection loss for long-term to suit the complex scenarios.

C. Unified Language-Based Evaluation

As LLMs become increasingly integrated into autonomous driving systems, the evaluation of generated textual outputs such as behavioral explanations, route descriptions, and visual question answering has emerged as a critical research concern. Conventional evaluation metrics, including BLEU, ROUGE, and METEOR [180], primarily rely on surface-level n-gram overlaps and are often insufficient for capturing the semantic fidelity or contextual appropriateness of the generated content in complex driving environments.

To address the limitations of traditional lexical metrics, semantically-aware methods have been adopted. BERTScore [181] uses contextual embeddings to better reflect human judgment. GPT-Score [182] leverages LLMs for evaluating fluency, relevance, and factuality, supporting both scalar scoring and preference ranking. CIDEr [183] and SPICE [184], though designed for image captioning, are applied to structured outputs like scene descriptions. For driving-specific tasks, metrics such as action accuracy and risk identification precision evaluate the alignment between language outputs and actual vehicle behavior or perceived scene risks [185].

VII. BENCHMARK RESEARCH

Benchmark studies are crucial for advancing models and optimizing dataset usage. However, given the vast diversity in autonomous driving scenarios and sensor configurations, evaluating algorithms solely based on performance on a single dataset can be misleading. Therefore, our aim is to analyze how variations in scene characteristics lead to algorithmic biases, rather than to provide a definitive performance ranking.

A. Perception

In map-based localization, NetVLAD [186] and Point-NetVLAD [187] are widely used as the foundational methods for tasks like place recognition and image geolocalization. Additionally, several methods such as BEV-based approaches [188], Transformer-based techniques [189], intruding occupancy maps [190], uncertainty estimation [191], and cross-modality-based methods [192] have emerged and achieved competitive results in map-based localization. Regarding the matching method in visual localization, LOL [193] and GSV [194] can be considered as the fundamental approaches. Furthermore, the performance of matching structures can be enhanced by utilizing siamese networks [195], attention mechanisms [196], and Generative Adversarial Networks [197]. Numerous reference links and evaluations can be made based on data from different scenes on SLAM methods¹. KITTI [5], UTBM [26], Oxford [8] are commonly used as the validation datasets. Introducing additional units into LiDAR-based methods like [198], [199] achieve SOTA results in these datasets.

¹ <https://github.com/younggunchu/awesome-slam-datasets>

And learning-based methods achieve competitive results in visual SLAM and visual localization. Compared to other tasks, localization is more closely tied to the specific contexts in which it is used. Consequently, it is common for users to provide proprietary data as benchmarks, especially for unique scenarios such as GPS signal loss or tunnel environments [200].

The distinction between segmentation and detection has become increasingly blurred in network architectures. Although tasks were categorized separately earlier, differentiating methods based on visual and LiDAR data is more appropriate when discussing benchmark performance. Transformer-based [201] and large-model approaches [202] now dominate in tasks like traffic sign detection, drivable area recognition, lane marking detection, semantic and instance segmentation, and pedestrian detection. Current research is shifting towards fine-grained reconstruction, cross-dataset generalization [203], and unsupervised training [204]. BDD-100K [12], Cityscape [18], KITTI [5] and Mapillary [77] are commonly used in visual detection and segmentation. In line with general visual tasks, testing data in autonomous driving scenarios commonly involves classic models such as YOLOP [205], DeepLabv3+ [206], and DETR [207]. However, SOTA performance is predominantly achieved by Transformer-based models [208], [209], and more recently, models built on LLM architectures [210], [211] have shown even more promising results. Currently, researchers generally treat autonomous driving scenarios as a specific case within visual tasks, where approaches such as cascades [212] and novel loss functions [213] yield strong results. In the era of large models, managing data effectively and using richer annotations or prompts have become even more critical for advancing performance. Transformer-based methods [214], [215] achieve SOTA performance in point segmentation task. Due to the inherent challenges in describing and representing 3D data alongside textual information, current research on point clouds using LLMs lags behind advancements in visual data processing. Besides, Transformer-based methods still obtain competitive results in BEV [216] and occupancy [217] tasks. For disparity, flow, and depth estimation, accurate matching [218], additional condition [219] can develop the performance. However, due to the rising popularity of surround-view systems and fusion strategies, interest in these traditional vision-based tasks is gradually declining. KITTI [5], Waymo [6], and nuScenes [7] are commonly selected for tracking task, and abandoning traditional detection-tracking approaches, Transformer-based methods [220] have now become the mainstream in the field. Prediction approaches can achieve better results by utilizing more historical data [221], making better decision choices [222], and employing Transformer-based [223] architectures.

B. Planning & End-to-End

The planning module can be divided into three main components, including route planning (start and end points), behavioral decision-making (execution strategy), and motion planning (motion trajectory). Common planning strategies are consistent with research in the field of mobile robotics, including Dijkstra [224] and A* [225], POMDP [226] and PCB [227],

Hybrid A* and RRT [228], etc. UniAD [229], GameFormer [230] prove that the Transformer-based structure can also achieve satisfactory results for this task in real data or virtual platforms, and unlike perception tasks, planning is more dependent on the specific scenario. The method can be evaluated from multiple perspectives, including error, time, efficiency, safety, and interpretability [231]. For the end-to-end task, research topics include behavior cloning, inverse optimal control, and reinforcement learning. Evaluation standards for these approaches are evolving, with considerations for safety, interpretability, multi-task learning, and causal confusion, among other factors [232].

C. LLMs

Visual question answering and cross-modal aggregation have long been explored in autonomous driving. With the emergence of LLMs, these tasks, now part of the broader domain of vision language models (VLMs), are gaining renewed attention. Unlike earlier works that relied on limited or manually curated datasets, large-scale benchmarks such as nuScenes [7] and Waymo [6] now include extensive language annotations, facilitating multimodal learning and evaluation [233]. Models such as DriveLM [234], DriveMLM [235], LMDrive [236], and DriveMM [237] have demonstrated promising performance across language-guided driving tasks. In parallel, a growing body of benchmarks has emerged to evaluate LLMs' performance in autonomous driving from diverse perspectives. For example, LingoQA [238] and Reason2Drive [239] evaluate VQA accuracy and reasoning consistency via GPT-4 alignment and human preferences. LaMPilot [240] and DriveSim [241] test instruction execution and real-world understanding in simulation, whereas OmniDrive [242] and AIDE [243] benchmark advanced capabilities such as 3D grounding and automated data engine generation. Together, these benchmarks underscore the increasing emphasis on semantic accuracy, safety, and generalization as critical criteria in evaluating language-integrated autonomous driving systems [244].

VIII. FUTURE RESEARCH

In this chapter, based on the previous analysis, we explore potential future research directions from a data-centric perspective.

1) *Data Optimization and Corner Cases*: Secondary annotation (or reprocessing) of existing datasets [127], [128] and the release of new datasets play a vital role in enhancing the research community's resources. Secondary annotation revitalizes original data by improving its relevance and accuracy, while new datasets broaden the diversity of training and testing scenarios. Alongside these efforts, identifying and properly handling corner cases is critical for vehicle safety and a defining challenge in autonomous driving. Although simulation platforms can generate corner cases, real-world corner case datasets [245] remain invaluable. Further work is needed to optimize the utilization of such datasets, integrate them with other driving conditions, and develop robust evaluation metrics. Providing stable evaluation tools and clear usage guidelines, as well as attention to data licenses, is essential to support these initiatives.

2) *Standardization and Data Loop*: Standardized data annotation and descriptions ensure consistency across various sources, improve data quality, and enhance model generalizability, while simplifying model evaluation and comparison. Unified annotation standards foster data sharing and reuse, reduce redundant labeling costs, and enable scalable, automated annotation, accelerating the development of autonomous driving technologies. The "data loop" concept refers to a feedback mechanism where perception and control data from autonomous vehicles continuously feed back into the system for ongoing learning and optimization [246]. This loop operates in offline and online modes, supporting model refinement and real-time adaptive decision-making. Exploring data loops not only improves system performance but also facilitates data commercialization by creating value through continuous data collection, refinement, and application in diverse scenarios.

3) *Legality, Ethics, and Privacy*: Ethical and legal considerations remain central to the deployment of autonomous driving systems. Embedding ethical guidelines and legal requirements at the data level can significantly influence driving decisions, ensuring alignment with societal norms and regulatory frameworks. This integration helps mitigate conflicts in edge cases, such as accident scenarios, while fostering public trust and ensuring compliance with regional laws. Privacy protection must also be rigorously addressed to maintain user confidence and meet legal obligations.

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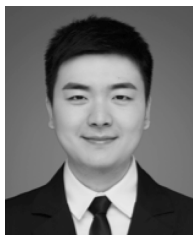
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